

Hana Ahmed Sakr

Contact no.: 01011562655

Email address: hanaahmedsagr@gmail.com

Address: Maadi , Cairo.

Linkedin: <https://www.linkedin.com/in/hana-sagr-0a119a3b2/>

Personal Summary

Results-driven third year Media (computer) Engineering student with strong analytical and problem-solving abilities. Passionate about building real-world tech solutions and mastering new technologies, with an eagerness to add value to a collaborative team.

Education

German University in Cairo (GUC) B.Sc. in Media Engineering [2023-2028]

Projects

1. **Enterprise HR Management System | 2025** (Engineered a full-stack web application integrated with a custom SQL Server database to manage university-wide HR operations using procedures, views, and functions)
2. **Advanced RAG Evaluation & Tuning Pipeline | 2026** (Engineered a hallucination-free RAG pipeline in Python, achieving perfect 1.0 Context Relevance scores by optimizing source text chunking and building custom automated evaluation scripts powered by the Gemini API)
3. **Software Engineering Student Portfolio Platform | 2026** (Engineered the front-end architecture for a scalable student portfolio platform, defining and implementing 30+ functional and non-functional user stories. Developed a core 'Project Roadmap' module to map milestones and utilized GitHub to manage version control and collaborative workflows)
4. **Operating Systems Resource Manager | 2026** (Architected robust operating system logic—including system calls, process states, memory, and a custom scheduler—to optimize CPU performance. Implemented strict synchronization using mutexes to manage concurrent resource allocation and completely eliminate deadlocks)
5. **Java Game Development: Jackaroo | 2025** (Developed a Java adaptation of the Jackaroo board game for one human player against 3 automated CPUs. Programmed game logic, including track navigation, collision detection and actions)
6. **Railway Network Optimizer | 2025** (Collaborated within a four-person team in Haskell to model stations and traversal costs. Implemented Dijkstra's algorithm and a recursive Priority Queue data structure to calculate shortest path between stations)

Technical Skills

- **Programming Languages:** Java, Python, Haskell, Prolog, JavaScript, VHDL
- **Databases:** SQL (SQL Server), Query Optimization
- **Tools:** React, HTML, CSS, Git/GitHub, API Developer , RAG & GIMP

Extracurricular Activities

- **Applied AI Committee Member | IEEE GUC Student Branch**
- **Associates Experience Committee Member | TEDxGUC**
- **Delegate | GUC MUN**
- **Usher | GUC SCAD Employment Fair**
- **Usher | GUC orientation**

Soft Skills

- Public Speaking & Presentation
- Team Work & Collaboration
- Time Management